

The information
is fast.

The portfolio
decision is *not*.

FinSense-AI

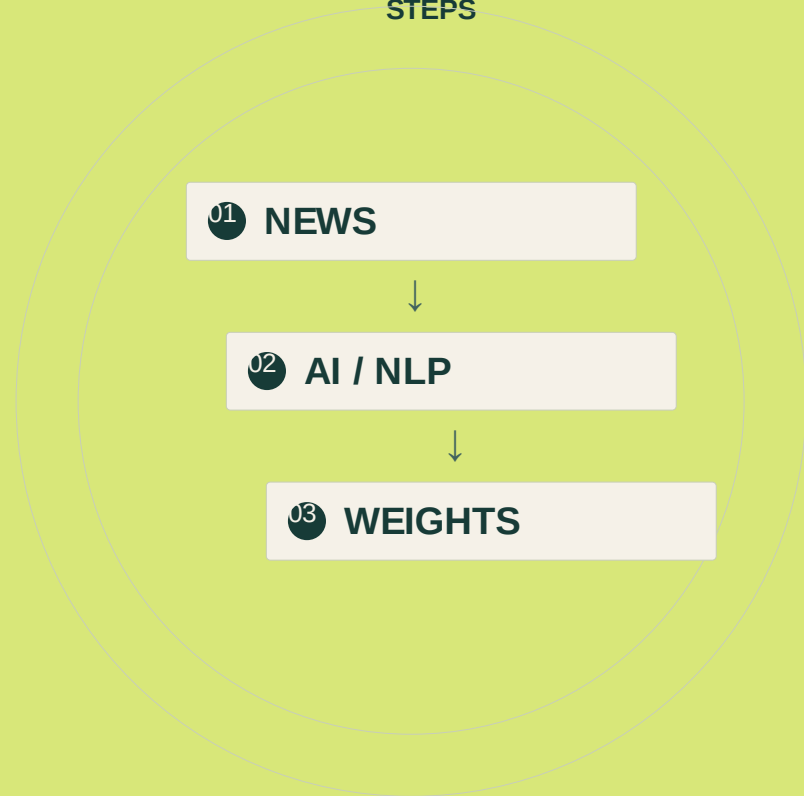
Tactical Stock Index Rebalancer

A decision engine that converts financial news
into bounded, explainable index-weight
changes.

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ONE SIGNAL / THREE
STEPS



FROM HEADLINES · TO BOUNDED DECISIONS

01/ 07

01 / PROBLEM & APPROACH

The information is fast.
The portfolio decision is *not*.

REBALANCE LATENCY

Review lag

Reviews can lag fast news.

INFORMATION OVERLOAD

Headline load

Manual news triage is slow.

EXECUTION DRAG

Signal churn

Reactive signals can raise turnover.

THE APPROACH

01

Ingest

GDELT +
PhraseBank

02

Interpret

FinBERT +
event scores

03

Fuse

Combine news
signals

04

Constrain

Water-fill · 2–
15% limits

05

Evaluate

Equal-weight
check

15-stock mock universe · 2–15% per stock · 100% portfolio budget

02 / SYSTEM DESIGN

News in. Bounded weights out.

01 / INGEST

Raw news

GDELT 2.0
PhraseBank
Deduplicate news

02 / SCORE

FinBERT

Sentiment: -1 to +1
10 event classes
Impact: 1-10

03 / FUSE

Signal fusion

Weighted inputs
Momentum

04 / ALLOCATE

Water-filling

15 mock stocks
Floor 2% · cap 15%
Total = 100%

SIGNAL WEIGHTS · 0.50 sentiment + 0.30 normalized impact + 0.20 momentum

SERVING & EXPLANATION

FastAPI → React 19 · SQLite WAL fallback · PostgreSQL-ready

03 / EVALUATION SETUP

Set the bounds. Measure the outcome.

PROTOTYPE SETUP

15

mock index stocks

2–15%

per-name floor / cap

100%

portfolio budget

6.67%

equal-weight baseline

10 bps

transaction-cost
assumption

71 / 71

automated tests passed

Point-in-time evaluation avoids look-ahead. The supplied PDF does not state the backtest period.

04 / KEY RESULTS

Prototype gains vs equal weight.

SOURCE-REPORTED PROTOTYPE BACKTEST

+3.82%

Excess return

Cumulative · net of 10 bps vs
equal weight

1.74

Sharpe ratio

vs 1.21 equal-weight baseline

2.40%

Drawdown reduction

as reported in source PDF

42%lower turnover vs naive
rebalancing**71/71**

automated tests passed

Net of 10 bps; 15-stock mock universe. Period not stated in source. Prototype backtest only—not realized investment performance.

05 / DOMAIN IMPACT

A clearer signal-to-decision hand-off.

Potential value for index teams—not an automated mandate.

01 RESPONSIVENESS

Earlier review

Surface event-driven proposals between scheduled index reviews.

03 RISK CONTROL

Bounded weights

Keep each stock within a 2–15% band and the portfolio at 100%.

02 EXPLAINABILITY

Clear attribution

Trace each tilt to the headline, sentiment, event class and score.

04 ACCESS

Open inputs

Public feeds and open tools may reduce paid-API dependence.

Potential impact—not evidence of production compliance or live-market alpha.

06 / LIMITS & NEXT STEPS

Validate before scaling up.

CURRENT LIMITS

15-stock mock universe

English headlines and summaries only

GDELT polls on a 15-minute cycle

Evaluation period not specified

NEXT STEPS

Expand index constituents

Add VaR / CVaR and stress tests

Test filings and multilingual news

Validate across market regimes

Thank you.

Questions?